

| Question | Answer                                                                                                              | Marks     | Guidance                                                                                                                                                                                                                                                                                                                          |
|----------|---------------------------------------------------------------------------------------------------------------------|-----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 3(a)     | sum of potential energy and kinetic energy                                                                          | <b>B1</b> | <b>Allow</b> 'PE' or ' $E_P$ ' for 'potential energy'.<br><b>Allow</b> 'KE' or ' $E_K$ ' for 'kinetic energy'.                                                                                                                                                                                                                    |
|          | (total) energy of random motion of particles                                                                        | <b>B1</b> | <b>Allow</b> 'distribution' for 'motion'.<br><b>Allow</b> 'random energy of particles'.<br><b>Allow</b> 'atoms' or 'molecules' for 'particles'.<br><b>Do not allow</b> 'system' or 'gas' for 'particles'.<br><b>Do not allow</b> 'a particle' for 'particles'.<br><br><b>Ignore</b> any reference to first law of thermodynamics. |
| 3(b)(i)  | $pV = NkT$                                                                                                          | <b>C1</b> | <b>Allow</b> $pV = nRT$ <b>and</b> $N = nN_A$ .                                                                                                                                                                                                                                                                                   |
|          | $V = (8.7 \times 10^{23} \times 1.38 \times 10^{-23} \times 300) / (2.0 \times 10^5) = 0.018 \text{ m}^3$           | <b>A1</b> | Full substitution and answer needed.<br>Unit not needed but any unit given must be correct.<br><b>Allow</b> '8.31 / $6.02 \times 10^{23}$ ' for ' $1.38 \times 10^{-23}$ '.                                                                                                                                                       |
| 3(b)(ii) | potential energy of <u>molecules</u> is zero                                                                        | <b>B1</b> | <b>Allow</b> ' $E_P$ ' or 'PE' for 'potential energy'.<br><b>Allow</b> 'PE = 0 because there are no intermolecular forces'                                                                                                                                                                                                        |
|          | internal energy = $\left(\frac{3}{2}\right) \times NkT$                                                             | <b>C1</b> | <b>Allow</b> internal energy = $\left(\frac{3}{2}\right) pV$ .                                                                                                                                                                                                                                                                    |
|          | $= \left(\frac{3}{2}\right) \times 8.7 \times 10^{23} \times 1.38 \times 10^{-23} \times 300$<br>$= 5400 \text{ J}$ | <b>A1</b> | Correct to at least 2 significant figures. (5403).<br><b>Allow</b> 5.4 kJ if unit is clear.                                                                                                                                                                                                                                       |

| Question | Answer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Marks | Guidance                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
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| 3(c)     | <p><u>Any three points from:</u></p> <p><u>Up to two points relating to change AB</u></p> <ul style="list-style-type: none"> <li>no change in internal energy</li> <li>work done = thermal energy transferred</li> <li>work done on the gas <b>and</b> thermal energy transferred from it</li> </ul> <p><u>Up to two points relating to change BC</u></p> <ul style="list-style-type: none"> <li>internal energy is halved</li> <li>no work done (on/by gas)</li> <li>decrease in internal energy = thermal energy removed</li> </ul> <p><u>Up to two points relating to change CA</u></p> <ul style="list-style-type: none"> <li>internal energy is doubled</li> <li>work done by gas = 1.8 kJ</li> <li>thermal energy transferred to gas = 4.5 kJ</li> </ul> <p><u>Up to two points relating to the complete cycle ABCA</u></p> <ul style="list-style-type: none"> <li>no overall change in internal energy</li> <li>overall work is done on gas <b>and</b> thermal energy is removed from it</li> <li>net work done = net thermal energy transferred</li> </ul> | B3    | <p><u>List Rules apply.</u></p> <p>All numerical values must be given with a unit and be correct to at least 2 significant figures.</p> <p><b>Ignore</b> points that are simply descriptions of or direct read-offs from the graph.</p> <p><b>Ignore</b> working space. No ECF.</p> <p>Throughout: Must be clear which aspect of the cycle each point is being made in relation to.</p> <p><b>Ignore</b> points that are not related to internal energy, thermal energy or work done</p> <p><b>Ignore</b> symbols unless defined</p> <p><b>Allow</b> 'heat' for 'thermal energy'.</p> <p><b>Allow</b> 'internal energy decreases' unless quantified as a factor different than halving.</p> <p><b>Allow</b> internal energy becomes 2.7 kJ.</p> <p><b>Allow</b> thermal energy removed = 2.7 kJ.</p> <p><b>Allow</b> 'internal energy increases' unless quantified as a factor different than two.</p> <p><b>Allow</b> 'internal energy becomes 5.4 kJ / increases by 2.7 kJ'</p> |